

STATEMENT OF PURPOSE

Applicant: Xuan Thanh Trinh

Program: Ph.D. in Electrical, Electronics and Computer Engineering

Institution: University of Michigan-Dearborn

Introduction

As a final-year Electrical Engineering student at Hanoi University of Science and Technology (HUST), my research at the Power Grid and Renewable Energy Laboratory focuses on distributed mathematical optimization for energy management. While this foundation drove my thesis on microgrid resilience, I recognize that classical optimization faces critical limitations when scaling to the real-time demands of modern grids. To overcome these bottlenecks, I seek to deepen my expertise in scalable cyber-physical control systems. UM-Dearborn's Ph.D. program provides the exact framework needed to architect decentralized grids for high renewable penetrations.

Academic Background

Ranking top 3 in HUST's Advanced Program in Electrical Engineering instilled in me a systems-level perspective, training me to deconstruct complex power networks into dynamic, tractable optimization problems. This mathematical foundation drives my independent research trajectory, allowing me to architect solutions that bridge abstract algorithms with real-world physical constraints.

In research submitted to *Sustainable Energy, Grids and Networks* (SEGAN), I developed a peer-to-peer trading framework that subdivides distribution networks into autonomous microgrids, integrating rigorous grid physics (AC-OPF) with distributed optimization (ADMM). I validated this architecture across both radial and ring-connected topologies, proving it reduced operational costs while strictly bounding voltage levels within the $\pm 5\%$ threshold and preventing local trading from destabilizing the public grid. However, this static framework lacked temporal foresight to handle sudden failures.

To enhance grid resilience and achieve dynamic fault tolerance, my graduation thesis layered Model Predictive Control (MPC) over this baseline. By coordinating emergency power exchanges during simulated outages, this temporal control empowered neighboring networks to assist a failing grid, reducing forced load shedding from 15 MWh to just 3 MWh while protecting 100% of critical loads.

Overcoming strict convergence challenges and collaborating across research groups forged my resilient analytical foundation. Equipped with this physics-based modeling capability, I am prepared to drive scalable, real-time grid operations within your doctoral ecosystem.

Research Interests

My primary research objective centers on the comprehensive optimization of autonomous power systems. I focus on advancing economic dispatch and Unit Commitment (UC) for daily operational efficiency, while engineering dynamic network reconfiguration and rapid fault response for high survivability against extreme disruptions. These deterministic models guarantee strict operational constraints across both routine economic scheduling and critical self-healing interventions.

My secondary research integrates Artificial Intelligence to complement traditional grid operations. My motivation lies in uncovering the untapped capabilities that emerge when fusing data-driven algorithms with physical models. To investigate this synergy, I target two Multi-agent Reinforcement Learning (MARL) applications: acting as a computational accelerator for instantaneous decentralized dispatch, and mitigating SOCP exactness issues alongside static thermal constraints.

Academic and Research Alignment

The University of Michigan-Dearborn provides the rigorous theoretical coursework and collaborative research environment required to pursue my synergistic vision. Specifically, my trajectory aligns directly with the

laboratory of Prof. Van-Hai Bui, driven by a shared focus on autonomous multi-agent energy management. Equipped with my deterministic optimization background, I aim to contribute to his learning-based frameworks, such as FairMarket-RL and XRL-LLM, by embedding my mathematical models as physics-informed safety filters.

Beyond Prof. Bui's laboratory, the expertise of Dr. Wencong Su and Dr. Junho Hong strongly complements my holistic research vision. Dr. Su's research on utilizing LLMs for Economic Dispatch synergizes perfectly with my focus on system-level AI integration and large-scale energy management. Concurrently, Dr. Hong's expertise in physics-informed anomaly detection directly reinforces my approach to component-level physical constraints and grid security. Fusing these perspectives expands my research horizon, providing the cross-disciplinary insights necessary to engineer highly resilient, autonomous grid architectures.

Career Objectives

Following the completion of my Ph.D, I intend to pursue a career as an R&D scientist dedicated to the advancement of global power infrastructures. I am deeply committed to applying these theoretical breakthroughs to solve high-impact challenges in developing regions, specifically driving the technical modernization and resilience of Vietnam's national grid.